从 SnapMirror 目标卷还原文件

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从 SnapMirror 目标卷还原文件

从 SnapMirror 目标还原单个文件， LUN 或 NVMe 命名空间

您可以从 Snapshot 副本还原单个文件， LUN ，一组文件或 LUN ，也可以从 SnapMirror 目标卷还原 NVMe 命名空间。从 ONTAP 9.7 开始，您还可以从 SnapMirror 同步目标还原 NVMe 命名空间。您可以将文件还原到原始源卷或其他卷。

要从 SnapMirror 同步目标（从 ONTAP 9.5 开始受支持）还原文件或 LUN ，必须先删除并释放此关系。

要将文件或 LUN 还原到的卷（目标卷）必须为读写卷：

- 如果源卷和目标卷具有通用 Snapshot 副本（通常情况下，还原到原始源卷时）， SnapMirror 将执行 _incremental restore" 。
- 否则， SnapMirror 将执行 _baseline restore" ，其中指定的 Snapshot 副本及其引用的所有数据块将传输到目标卷。

步骤

1. 列出目标卷中的 Snapshot 副本：

   `volume snapshot show -vserver` *`svm`* `-volume volume`

   有关完整的命令语法，请参见手册页。

   以下示例显示了 `vserverB` ： `secondary1` 目标上的 Snapshot 副本：

```
cluster_dst::> volume snapshot show -vserver vserverB -volume secondary1

Vserver     Volume      Snapshot                State    Size  Total%
Used%
-------     ------      ---------- ----------- ------   -----  ------
-----
vserverB    secondary1  hourly.2013-01-25_0005  valid   224KB     0%
0%
                        daily.2013-01-25_0010   valid   92KB      0%
0%
                        hourly.2013-01-25_0105  valid   228KB     0%
0%
                        hourly.2013-01-25_0205  valid   236KB     0%
0%
                        hourly.2013-01-25_0305  valid   244KB     0%
0%
                        hourly.2013-01-25_0405  valid   244KB     0%
0%
                        hourly.2013-01-25_0505  valid   244KB     0%
0%

7 entries were displayed.
```

2. 从 SnapMirror 目标卷中的 Snapshot 副本还原单个文件或 LUN 或一组文件或 LUN：

   `snapmirror restore -source-path` *`SVM：volume|cluster：//svm/volume，…`* `-destination-path` *`svm：volume|cluster：//svm/volume，…`* `-source-snapshot snapshot -file-list` *`source_file_path，@destination_file_path`*

   有关完整的命令语法，请参见手册页。

   > 您必须从目标 SVM 或目标集群运行此命令。

   以下命令会将文件 `file1` 和 `file2` 从原始目标卷 `secondary1` 中的 Snapshot 副本 `daily.2013-01-25_0010` 还原到原始源卷 `primary1` 的活动文件系统中的同一位置：

```
cluster_dst::> snapmirror restore -source-path vserverB:secondary1
-destination-path vserverA:primary1 -source-snapshot daily.2013-01-
25_0010 -file-list /dir1/file1,/dir2/file2

[Job 3479] Job is queued: snapmirror restore for the relationship with
destination vserverA:primary1
```

   以下命令会将文件 `file1` 和 `file2` 从原始目标卷 `secondary1` 中的 Snapshot 副本 `daily.2013-01-25_0010` 还原到原始源卷 `primary1` 的活动文件系统中的其他位置。

目标文件路径以 @ 符号开头，后跟原始源卷根中文件的路径。在此示例中， `file1` 还原到 `/dir1/file1.new` ， file2 还原到 `/dir2.new/file2` on `primary1` ：

```
cluster_dst::> snapmirror restore -source-path vserverB:secondary1
-destination-path vserverA:primary1 -source-snapshot daily.2013-01-
25_0010 -file-list
/dir/file1,@/dir1/file1.new,/dir2/file2,@/dir2.new/file2

[Job 3479] Job is queued: snapmirror restore for the relationship with
destination vserverA:primary1
```

以下命令会将文件 `file1` 和 `file3` 从初始目标卷 `secondary1` 中的 Snapshot 副本 `daily.2013-01-25_0010` 还原到初始源卷 `primary1` 的活动文件系统中的不同位置，并将 `file2` 从 `snap1` 还原到 **主** 文件系统中的同一位置。

在此示例中，文件 `file1` 还原为 `/dir1/file1.new` ，而 `file3` 还原为 `/dir3.new/file3` ：

```
cluster_dst::> snapmirror restore -source-path vserverB:secondary1
-destination-path vserverA:primary1 -source-snapshot daily.2013-01-
25_0010 -file-list
/dir/file1,@/dir1/file1.new,/dir2/file2,/dir3/file3,@/dir3.new/file3

[Job 3479] Job is queued: snapmirror restore for the relationship with
destination vserverA:primary1
```

从 SnapMirror 目标还原卷的内容

您可以从 SnapMirror 目标卷中的 Snapshot 副本还原整个卷的内容。您可以将卷的内容还原到原始源卷或其他卷。

还原操作的目标卷必须为以下卷之一：

- 读写卷，在这种情况下，如果源卷和目标卷具有通用 Snapshot 副本（通常情况下，还原到原始源卷时），SnapMirror 将执行 _incremental restore" 。

  > 如果没有通用 Snapshot 副本，则此命令将失败。您不能将卷的内容还原到空的读写卷。

- 空数据保护卷，在这种情况下， SnapMirror 会执行 _baseline restore" ，其中指定的 Snapshot 副本及其引用的所有数据块都会传输到源卷。

还原卷的内容会造成系统中断。还原操作正在运行时， SnapVault 主卷上不得运行 CIFS 流量。

如果还原操作的目标卷已启用数据压缩，而源卷未启用数据压缩，请在目标卷上禁用数据压缩。还原操作完成后，您需要重新启用数据压缩。

在执行还原之前，为目标卷定义的任何配额规则都将被停用。还原操作完成后，您可以使用 `volume quota modify` 命令重新激活配额规则。

步骤

1. 列出目标卷中的 Snapshot 副本：

   `volume snapshot show -vserver` *`svm`* `-volume` *`volum`*

   有关完整的命令语法，请参见手册页。

   以下示例显示了 `vserverB ： secondary1` 目标上的 Snapshot 副本：

```
cluster_dst::> volume snapshot show -vserver vserverB -volume secondary1

Vserver     Volume      Snapshot                State    Size  Total%
Used%
-------     ------      ---------- ----------- ------   -----  ------
-----
vserverB    secondary1  hourly.2013-01-25_0005  valid   224KB     0%
0%
                        daily.2013-01-25_0010   valid   92KB      0%
0%
                        hourly.2013-01-25_0105  valid   228KB     0%
0%
                        hourly.2013-01-25_0205  valid   236KB     0%
0%
                        hourly.2013-01-25_0305  valid   244KB     0%
0%
                        hourly.2013-01-25_0405  valid   244KB     0%
0%
                        hourly.2013-01-25_0505  valid   244KB     0%
0%

7 entries were displayed.
```

2. 从 SnapMirror 目标卷中的 Snapshot 副本还原卷的内容：

   `snapmirror restore -source-path` *`SVM`* `：` *`volume|cluster`* `：` *`//svm/volume`* `，` … `-destination-path` *`svm`* `：` *`volume|cluster`* `：` *`//svm/volume`* `，` … `-source-snapshot` *`snapshot`*

   有关完整的命令语法，请参见手册页。

   > 您必须从目标 SVM 或目标集群运行此命令。

   以下命令将从原始目标卷 `secondary1` 中的 Snapshot 副本 `daily.2013-01-25_0010` 还原原始源卷 `primary1` 的内容：

```
cluster_dst::> snapmirror restore -source-path vserverB:secondary1
-destination-path vserverA:primary1 -source-snapshot daily.2013-01-
25_0010

Warning: All data newer than Snapshot copy daily.2013-01-25_0010 on
volume vserverA:primary1 will be deleted.

Do you want to continue? {y|n}: y

[Job 34] Job is queued: snapmirror restore from source
vserverB:secondary1 for the snapshot daily.2013-01-25_0010.
```

3. 重新挂载已还原的卷并重新启动使用该卷的所有应用程序。

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